Omnivortex V1 — Phase 1 Session 1 Measurement Log

R=89mm r=55mm · 123T · 22 AWG · github.com/Te55er4ct/phi-harmonic-toroidal-field

Date:	**06/06/2026**	**Time:**	**15:03-16:44 CDT**	**Temp:**	**84°F outdoor**	**Humidity:**	**38%**	**Location:**	**Outdoors / wooden sawhorse**

Section 1 — LCR Meter (Proster BM4070)

Fixed freq ~1kHz

Parameter	~1kHz	10kHz	100kHz	Units	Notes
Inductance L	**0.47**	—	—	mH	Confirmed indoors & outdoors
Q factor	**~1.0**	—	—	—	Calc: wL/R; NanoVNA: 161@500k, 14@SRF
ESR (R_s)	—	—	—	ohm	Not displayed on BM4070
DC resistance	**2.9**	—	—	ohm	Ames CM610A clamp meter

Section 2 — NanoVNA-H Self-Resonant Frequency

fw v1.2.44 / Saver 0.7.3

Measurement	Value	Units	Notes
SRF primary peak	**~1.000**	MHz	Phase zero crossing 1.000-1.00040 MHz
SRF harmonic 2	—	MHz	Not yet measured
SRF harmonic 3	—	MHz	Not yet measured
C_self calculated	**~54**	pF	C=1/(4pi^2 x f^2 x L)

Section 3 — Impedance Sweep Key Points (500kHz - 30MHz)

Feature	Freq (MHz)	|Z| (ohm)	Phase (deg)	Notes
Inductive region low end	**0.500**	**~1880**	**+3.04**	Series L ~600uH, Q=161
SRF primary	**~1.000**	**~10500**	**~0**	Phase zero crossing confirmed
Q at SRF	**~1.000**	**~10500**	**~0**	Q = ~14 near SRF
Capacitive region start	**>1.001**	—	**neg**	Series L goes negative
VSWR minimum (wide)	**12.005**	**141**	**-20.53**	Min VSWR 6.452
Anti-resonance	**12.005**	**141**	**-20.53**	Series L = -2.06uH
Upper limit	**30.000**	—	—	No further features this session

Section 4 — Instrument & File Log

Cal file:	V1_500k_30M (OSL via calibration assistant)
.s1p file:	V1_SRF_1M.s1p
Screenshots:	nanovna_2026-06-06_16-12-32 thru 16-44-31 (8 captures)
NanoVNA fw:	NanoVNA-H v1.2.44
Instruments:	Proster BM4070 LCR | Ames CM610A clamp | NanoVNA-H

Observations / anomalies / notes

L=0.47mH confirmed identical indoors and outdoors — no EMI effect from indoor environment on LCR reading.
SRF ~1.000MHz by phase zero crossing. Capacitive above, inductive below. C_self ~54pF calculated.
NanoVNA SOL cal failed at 100k and 500k start frequencies. Resolved via calibration assistant at 500k.
Connection: bare coil wire wrapped on SMA connector. SMA-to-alligator pigtails needed for Session 2.
DC resistance 2.9 ohm (spec 3.0 ohm). WiFi active on laptop — no artifacts visible in sweep range.

Operator:	**Wierzbicki**	**Session #:**	**1**	**Git commit hash:**	**(fill after push)**